Lab 2: Object Oriented Programming

Parent Class

Create a parent class ComplexNumber having two protected data members:
- _real (float)
- _imaginary (float)

- Tasks:

1. Create the following constructors:
    - Default constructor: assigns -1 to both members.
    - Parameterized constructor: takes real and imaginary values as arguments.
    - Copy constructor equivalent: takes another ComplexNumber object and copies its values.
2. Write a method display() that prints the number in the form a+ib or a-ib.
    - Handle negative imaginary parts correctly.

Child Class 1

Create a child class ComplexArithmetic which inherits from ComplexNumber.

Tasks:

3. **Constructors**:
    - Default constructor.
    - Parameterized constructor.
4. **Methods**:
    - add(self, other) → returns a new ComplexArithmetic object which is the sum of two complex numbers.
    - multiply(self, other) → returns a new ComplexArithmetic object which is the product of two complex numbers.
5. **Destructor**: print a message when the object is destroyed.

Child Class 2

Create a child class ComplexAdvanced which inherits from ComplexArithmetic.

Tasks:

6. Constructor: accepts real and imaginary parts.
7. Methods:

- ***conjugate(self)*** → returns a new ComplexAdvanced object which is the conjugate of the complex number.
- ***modulus(self)*** → returns the magnitude of the complex number, given by:

  $|z| = \sqrt{(real^2 + imaginary^2)}$

8. Destructor: print a message when the object is destroyed.

Main Program (Example Driver)

- Write a driver program (if __name__ == '__main__':) that:

9. Creates ComplexNumber objects using default, parameterized, and copy constructors and displays them.
10. Creates two ComplexArithmetic objects, adds them, multiplies them, and displays results.
11. Creates a ComplexAdvanced object, computes its conjugate and modulus, and displays results.